CIRC HQ STUDENT HANDOUT | GRADE 5

Upcycle Lab

Goal: Transform a clean discarded item into a useful prototype with a clearly defined user and function.

35 minutes | 8 steps | Engineering and Design | Playbook A meeting 17, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Demonstrate the new function and explain why it is more than decoration.

You will use

- Clean reclaimed objects
- Cardboard
- Tape
- String
- Fasteners
- Markers
- Safety scissors

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Choose one inspected reclaimed object.
- Place it on the project mat.

2 Step 2: Material safety — 3 min

- Use only the approved clean object.
- Cut only on the table.
- Keep fingers outside the cutting path.
- Carry scissors closed.

3 Step 3: Choose a new use — 5 min

- List two useful object properties.
- Name one user.
- Write one new function.

4 Step 4: Build the new use — 8 min

- Sketch the planned change.
- Cut only approved added material.
- Build the new function.

5 Step 5: Test the function — 6 min

- Run one clear use test.
- Record the result.
- Find one weak feature.

6 Step 6: Improve the use — 5 min

- Change the weak feature.
- Repeat the use test.
- Prepare a before-and-after explanation.

7 Step 7: Reset the lab — 2 min

- Close the scissors.
- Sort every unused material.

8 Step 8: Exit check — 3 min

- Name the reused property.
- State the new function.

What to turn in

Your object's new job, explained before and after.

No computer? Another way to do it

Sketch the new use and explain it.

Finished early? Next challenge

Second Life Label: Create a simple label showing the old purpose, new purpose, and materials kept out of waste.

Stay safe

Use only inspected clean items with no sharp edges, residue, glass, batteries, or unknown contents.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.